

Canopy chlorophyll RSI, Rajanpur

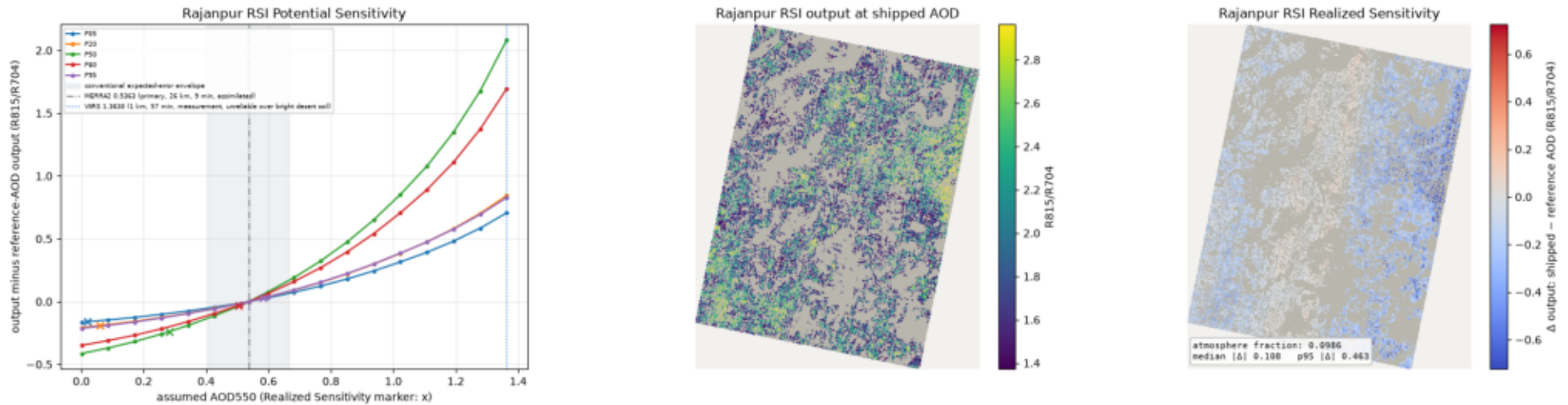


Figure 1. Canopy chlorophyll red-edge ratio, Rajanpur, Pakistan

Scene `20250114\_062056\_92\_4001`, acquired January 14, 2025. The Inoue et al. (2016) `R815 / R704` ratio over 211,873 canopy pixels, corrected at the shipped aerosol optical depth and at a MERRA-2 reference. Across the scene, 9.86 percent of the ratio's variance is the aerosol assumption rather than real canopy difference. The Realized Sensitivity map is almost all one sign, so the ratio shifts coherently across the whole scene rather than at scattered pixels. This is the most aerosol-exposed of the land-surface algorithms, which a ratio spanning a steep spectral slope predicts.

Wildfire SAM, Malibu

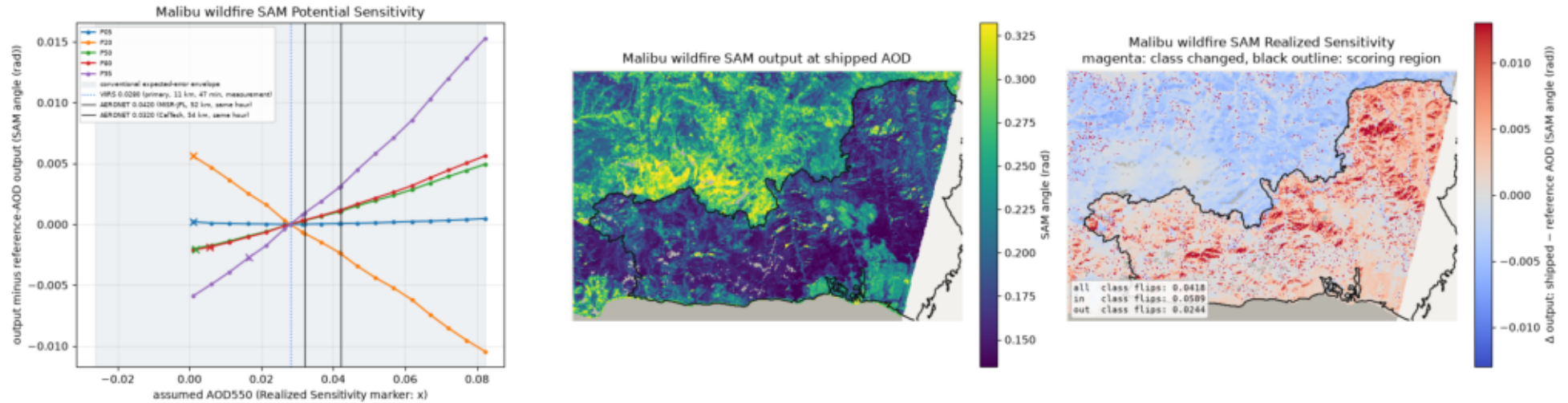


Figure 2. Wildfire spectral angle mapper, Palisades fire, Malibu, California

Scene `20250123\_185518\_92\_4001`, acquired January 23, 2025, days after the fire. A spectral angle mapper against a fixed 28 spectrum burned and unburned library, corrected at the shipped floor value and at a VIIRS reference. The magenta overlay marks pixels that change between burned and unburned, and the black outline is the WFIGS fire perimeter. The badge reports the flip fraction whole scene, inside the perimeter, and outside: 4.18 percent overall, 5.9 percent inside, 2.4 percent outside. Reassignment concentrates where surfaces are spectrally ambiguous, which is inside the burn.

Cyanobacteria index, Lake Ontario

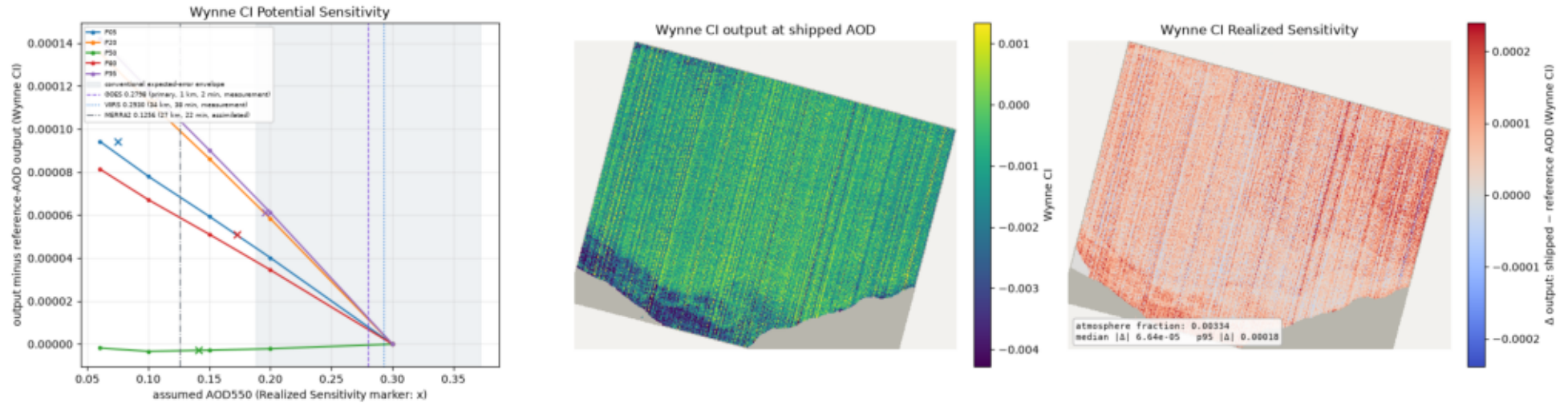


Figure 3. NOAA Cyanobacteria Index, western Lake Ontario

Scene `20250704\_165204\_61\_4001`, acquired July 4, 2025, over a bloom along the Rochester shoreline. The Wynne et al. (2008) Cyanobacteria Index, the curvature of the reflectance spectrum at 681 nm read as its value there minus a baseline from 665 and 709 nm, corrected at the shipped aerosol optical depth and at a GOES reference. Its variance fraction is 0.33 percent, the lowest in the set. Subtracting the baseline cancels a smooth spectral shift, so the index barely moves even though the aerosol error is large.

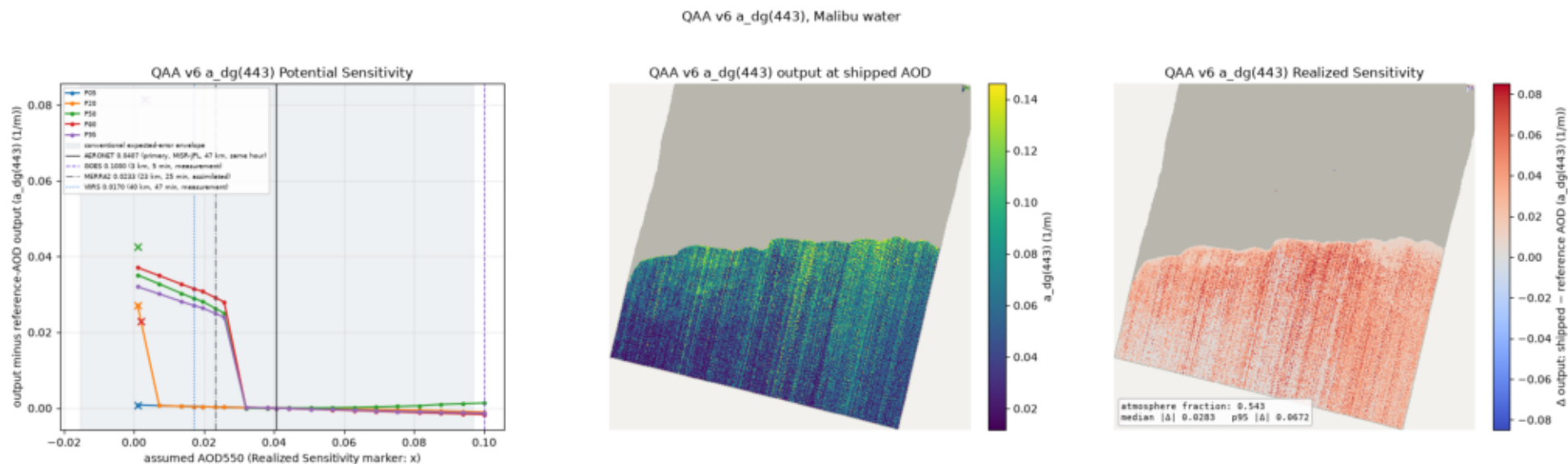


Figure 4. QAA v6 dissolved-and-detrital absorption, Malibu, California

Same scene as Figure 2, `20250123\_185518\_92\_4001`, over the coastal water. The QAA v6 semi-analytical inversion (Lee et al. 2002), corrected at the shipped floor value and at an AERONET reference. The panel shows `a\_dg(443)`, the dissolved-and-detrital absorption at 443 nm, whose variance fraction is 54 percent. Run on the same pixels the inversion also returns a phytoplankton absorption at 64 percent and a spectral slope at 2.7 percent. One algorithm produces both a near-immune output, the slope set by a ratio, and two of the most exposed outputs in the walkthrough, the absorption magnitudes.